

Stakeholders Audit

Wayfair delivers a visually polished shopping experience with strong accessibility, SEO, and a modern front-end architecture. However, the website suffers from significant client-side performance issues caused primarily by large JavaScript bundles and numerous third-party integrations.

These issues increase page load time, delay user interaction, and create a slower shopping experience, particularly for first-time visitors and users on slower devices. Poor performance can negatively affect customer engagement, conversion rates, and search visibility.

Not all improvements require major architectural changes. Several high-impact optimizations can be implemented with relatively low development effort, while a smaller number of structural improvements require longer-term investment.

Overall Assessment

Area	Assessment
Business Risk	Medium–High
User Experience	Moderate
SEO Impact	Moderate
Conversion Impact	High
Estimated Implementation Effort	Low–High (depends on recommendation)

What Is Working Well

Strong Search Engine Optimisation

The website already achieves an excellent SEO score and uses server-side rendering for the homepage, allowing search engines to index content efficiently.

Business value

- Better organic visibility
 - Lower customer acquisition cost
 - Faster discovery of products
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Modern Rendering Architecture

The homepage uses Server-Side Rendering followed by client-side hydration.

Business value

- Faster first visual load
 - Better reliability
 - Improved crawlability
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Good Accessibility

Accessibility is already implemented at a high level.

Business value

- Broader customer reach
 - Reduced legal risk
 - Better usability
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Highest Priority Opportunities

1. Reduce JavaScript Execution

Business Impact

Large JavaScript bundles significantly delay the time before customers can interact with the page.

This increases abandonment risk, particularly on mobile devices where CPU performance is lower.

Evidence

- Approximately **25 MB** of JavaScript downloaded.
- More than **5 seconds** spent executing JavaScript.
- Roughly **50%** of downloaded resources remain unused during initial page load.

Business Risk

High

Estimated Effort

Medium

Recommendation

Reduce initial JavaScript payload by:

- better code splitting
 - lazy loading
 - removing unused dependencies
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2. Reduce Third-Party Dependencies

Business Impact

Advertising, analytics, monitoring, and marketing scripts compete with shopping functionality for browser resources.

Every additional script increases loading time while offering diminishing business value.

Evidence

Heavy use of:

- Google Tag Manager
- Quantum Metric
- Facebook
- Stripe
- Signifyd

These collectively consume a significant portion of main-thread execution.

Business Risk

High

Estimated Effort

Low–Medium

Recommendation

Audit third-party services and defer or remove integrations that do not directly improve conversion or operational requirements.

3. Improve Initial Page Weight

Business Impact

Large downloads increase loading time for new visitors and customers on slower networks.

Slower pages correlate with lower conversion rates.

Evidence

- Hundreds of network requests
- Large overall download size
- Numerous CSS and JavaScript assets

Business Risk

Medium

Estimated Effort

Medium

Recommendation

Continue optimizing images, CSS delivery, caching strategy, and asset compression.

4. Improve Server Response Time

Business Impact

The initial server response exceeded **800 ms**, delaying the delivery of the first HTML document.

Although the rendering strategy is appropriate, reducing backend latency would improve perceived responsiveness.

Business Risk

Medium

Estimated Effort

Medium–High

Recommendation

Investigate server-side processing and caching opportunities to reduce Time to First Byte (TTFB).

Priority Matrix

Priority	Recommendation	Business Impact	Effort
1	Reduce JavaScript execution	Very High	Medium
2	Audit third-party scripts	High	Low–Medium
3	Reduce overall page weight	Medium	Medium
4	Improve server response time	Medium	Medium–High

Expected Business Benefits

Implementing these recommendations is expected to provide:

- Faster page loading and improved responsiveness.

- Higher customer satisfaction.
 - Reduced bounce rate.
 - Increased conversion potential.
 - Better mobile shopping experience.
 - Improved Core Web Vitals and long-term SEO performance.
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